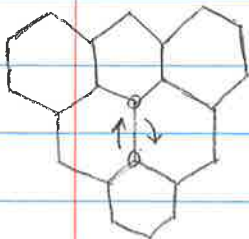


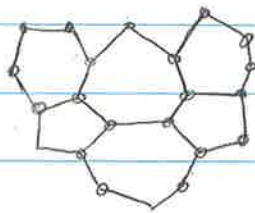
1. a) A unit cell of graphene is hexagonal in shape and made of six carbon atoms. In each unit cell, $\frac{1}{3}$ rd of these 6 carbon atoms are shared. Hence, carbon atoms per unit cell is;
 $\frac{1}{3} \times 6 = 2$ carbon atoms.

2. Stone-Wales (SW) defect refers to an intrinsic defect, i.e., crystalline order is perturbed without the presence of foreign atoms. In this defect, one of the C-C bonds in graphene is rotated by 90° which ultimately results in formation of 2 pentagons and 2 heptagons.

This rearrangement has important consequences in terms of chemical, mechanical and electrical properties of graphene. This is useful for some desired applications of graphene because all of graphene's potential properties cannot be accessed directly without any modifications.



C-C bond
by 90° rotation



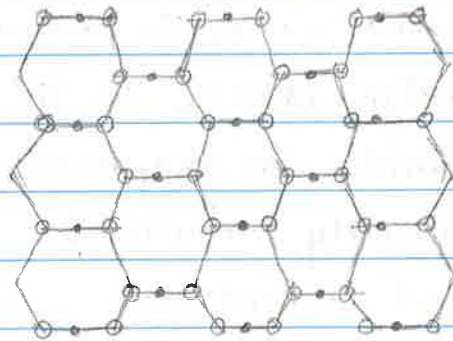
2 pentagons & 2 heptagons
are formed.

1. (b) i) lattice & its type

⇒ It is a hexagonal lattice

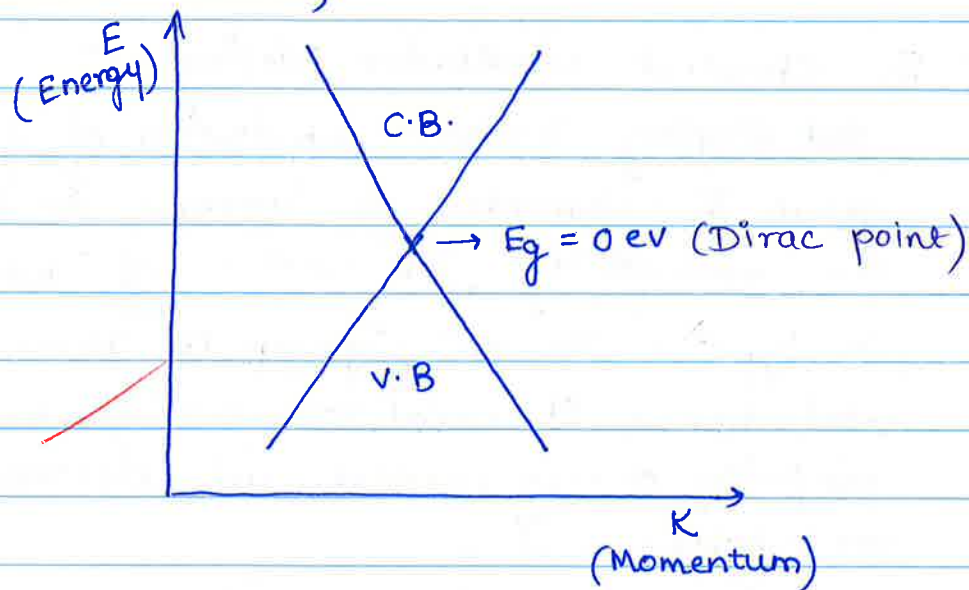
ii) Basis and how many atoms in basis, Crystal structure.

⇒ It has C-C bond as basis with 2 carbon atoms. It has a hexagonal crystal structure.



3. Intrinsic monolayer graphene is a zero bandgap semiconductor. Show the band structure (Energy-momentum linear dispersion) of an intrinsic monolayer graphene. Suggest some techniques (min. 2) to open bandgap in graphene.

⇒ E-K of an intrinsic monolayer graphene is as shown below;



Two techniques to open a band gap in graphene involves introduction of other compounds in the structural lattice of graphene and also the deposition of graphene on different substrates ~~to~~ show promising results. Many recent studies comprise of various ^{experimental} methods for opening band gap of graphene to utilize its semiconductor ~~to~~ properties. Two techniques for opening band gap in graphene are;

i) Substrate induced band gap opening:

This technique involves the deposition of graphene on substrates like h-BN or oxygen terminated SiO_2 . h-BN on Ni acts as an insulating substrate and for depositing single layer graphene which ultimately leads to opening of band gap by few eV.

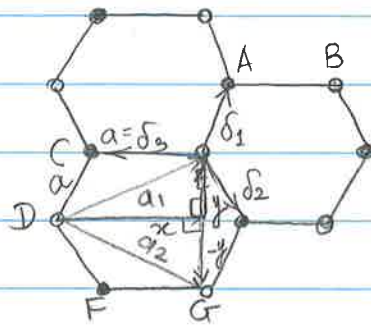
ii) Chemical substitution doping:

~~Sub~~ Doping involves substitution of group III or group V elements to increase or decrease the conductivity. It consists of two types;

n-type: In this group III elements are doped and hence it creates extra electrons, i.e., majority charge carriers are electrons and minority are holes.

p-type: In this group V elements are doped & it creates extra holes, i.e., majority carriers are holes & minority carriers are electrons.

4. Calculate the lattice unit vectors (a_1 & a_2) in the following honeycomb graphene crystal lattice.



⇒

Here length $AB = 0.142 \text{ nm}$

Then in $\triangle CDE$, $CD = CE = 0.142 \text{ nm}$

$DE = a_1$, $\angle C = 120^\circ$.

By law of cosines;

$$(DE)^2 = (CD)^2 + (CE)^2 - 2(CD)(CE)\cos C$$

$$a_1^2 = (0.142)^2 + (0.142)^2 - 2(0.142)^2 \cos 120^\circ$$

$$a_1^2 = 0.040328 + 0.020164$$

$$a_1^2 = 0.060492$$

$$a_1 = 0.246 \text{ nm}$$

Also here $a_1 = a_2 = 0.246 \text{ nm}$

Let; $\delta_1 = \delta_2 = \delta_3 = a$

Here from the fig,

$$a_1^2 = x^2 + y^2$$

By law of cosines; $a_1^2 = a^2 + a^2 - 2a^2 \cos 120^\circ$

$$a_1^2 = 2a^2 - a^2 = a^2$$

$$\therefore a^2 = x^2 + y^2$$

1. The first part of the problem is to find the area of the region bounded by the curve $y = \sqrt{x}$ and the line $y = x$.

2. The second part is to find the volume of the solid generated by revolving the region about the y-axis.

3. The third part is to find the volume of the solid generated by revolving the region about the x-axis.

4. The fourth part is to find the volume of the solid generated by revolving the region about the line $x = 2$.

5. The fifth part is to find the volume of the solid generated by revolving the region about the line $x = 4$.

6. The sixth part is to find the volume of the solid generated by revolving the region about the line $x = 6$.

7. The seventh part is to find the volume of the solid generated by revolving the region about the line $x = 8$.

8. The eighth part is to find the volume of the solid generated by revolving the region about the line $x = 10$.

9. The ninth part is to find the volume of the solid generated by revolving the region about the line $x = 12$.

10. The tenth part is to find the volume of the solid generated by revolving the region about the line $x = 14$.

11. The eleventh part is to find the volume of the solid generated by revolving the region about the line $x = 16$.

12. The twelfth part is to find the volume of the solid generated by revolving the region about the line $x = 18$.

13. The thirteenth part is to find the volume of the solid generated by revolving the region about the line $x = 20$.

14. The fourteenth part is to find the volume of the solid generated by revolving the region about the line $x = 22$.

15. The fifteenth part is to find the volume of the solid generated by revolving the region about the line $x = 24$.

16. The sixteenth part is to find the volume of the solid generated by revolving the region about the line $x = 26$.

17. The seventeenth part is to find the volume of the solid generated by revolving the region about the line $x = 28$.

18. The eighteenth part is to find the volume of the solid generated by revolving the region about the line $x = 30$.

19. The nineteenth part is to find the volume of the solid generated by revolving the region about the line $x = 32$.

20. The twentieth part is to find the volume of the solid generated by revolving the region about the line $x = 34$.

21. The twenty-first part is to find the volume of the solid generated by revolving the region about the line $x = 36$.

22. The twenty-second part is to find the volume of the solid generated by revolving the region about the line $x = 38$.

23. The twenty-third part is to find the volume of the solid generated by revolving the region about the line $x = 40$.

24. The twenty-fourth part is to find the volume of the solid generated by revolving the region about the line $x = 42$.